UNIVERSITY OF LIMPOPO

FACULTY OF SCIENCE AND AGRICULTURE
SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

Design Implementation and Secure Operation of a Multi Site Smart Campus Network

Final Project Report

Project detail	Information
Module	SCOB032 Computer Networks
Assessment	Semester Project
Group	GROUP NUMBER 10
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Submission date	9 October 2026

Declaration

We declare that this report represents the work completed for the SCOB032 Computer Networks semester project. All external sources, standards and technical documentation used in the design have been acknowledged in the reference list. Each member's contribution is recorded in the contribution statement at the end of this report.

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1 Executive Summary

This project designed and implemented a secure multi-site smart campus network for the University of Limpopo using Cisco Packet Tracer. The solution connects a main campus and a satellite campus and supports academic, administrative, residential and operational services. The design separates users and services through VLANs, uses variable-length subnetting to allocate address space efficiently, and provides redundant default gateways through HSRP. Two main-campus multilayer switches provide resilient inter-VLAN routing, while redundant access uplinks are controlled by spanning tree.

Security was implemented in layers. Extended access control lists enforce zero-trust-style restrictions between student, residence, research, IoT, management and server networks. Port security, BPDU Guard, unused-port shutdown, SSH management and wireless 802.1X reduce unauthorised access. An

IPsec site-to-site VPN protects traffic exchanged between the two campuses. Central DNS, DHCP, RADIUS, NTP, syslog, web and SDN services support the network.

The completed prototype supports wired and wireless connectivity, lightweight access points, staff and student authentication, IoT registration, VoIP calling and priority treatment for voice traffic. Testing confirmed DHCP allocation, VLAN reachability, ACL enforcement, HSRP failover, VPN encryption, wireless authentication, registered phones and inter-campus calls. Packet Tracer limitations affected detailed QoS counter visibility after IPsec encryption, but voice classification, DSCP marking and attached priority policies were verified before and at the WAN edge.

2 Introduction

2.1 Background

A modern university depends on reliable connectivity for teaching, research, administration, residences, security and digital services. Placing all users and devices in one broadcast domain would create avoidable congestion and expose sensitive systems to unnecessary risk. The proposed design therefore uses a hierarchical, segmented architecture that can serve more than 500 concurrent users while remaining manageable in a simulated environment.

2.2 Problem Statement

The university requires one coordinated network that connects multiple buildings at the main campus to a remote satellite campus. The network must carry data, voice, video, wireless and IoT traffic without allowing unrestricted movement between user groups. It must remain available during the failure of a gateway or uplink, protect traffic crossing the public network, provide centralised authentication and monitoring, and allow future expansion without redesigning the entire addressing plan.

2.3 Project Objectives

- Design logical and physical campus topologies using a scalable hierarchical model.
- Create VLAN and VLSM plans for users, voice, wireless, IoT, servers and management.
- Provide dynamic addressing and central services to both campuses.
- Implement resilient routing and switching using HSRP, redundant trunks and spanning tree.
- Secure the design using ACLs, zone-based firewalling, IPsec VPN, 802.1X, SSH and port hardening.
- Integrate central wireless control, SDN monitoring, IoT services, VoIP and QoS.
- Test connectivity, failover, isolation, authentication, services and performance.

2.4 Scope

The implementation covers the main campus buildings, including administration, lecture halls, library, student residences, mathematics staff and research laboratories, together with the server and ICT administration environment. The satellite campus contains staff, students, printers, voice, IoT, staff Wi-Fi, student Wi-Fi and management networks. Internet-scale routing was represented by an ISP segment, while Packet Tracer provided the routers, multilayer switches, access switches, controllers, access points, servers, phones, IoT devices and endpoints.

3 Requirements and Methodology

3.1 Requirements Analysis

Requirement	Implementation response
500+ concurrent users	Large student and residence subnets, VLAN segmentation and hierarchical switching
Multiple campuses	Encrypted site-to-site IPsec VPN over simulated ISP links
High availability	Dual main core switches, HSRP, redundant access uplinks and PVST
Secure access	ACLs, firewall zones, 802.1X/RADIUS, SSH and switch-port hardening
Wireless mobility	WLC-managed LAPs with staff, student and residence WLANs
Voice and video	CME voice gateways, separate voice VLANs and WAN QoS policies
IoT services	Dedicated IoT VLANs and central registration server
Management	Management VLAN, NTP, syslog, SDN controller and monitoring script

3.2 Implementation Method

The team followed an iterative build-and-test method. The addressing plan and VLAN structure were established first, followed by access-layer deployment and trunk verification. Layer 3 gateways, HSRP and DHCP relay were then configured. Security controls were introduced after basic reachability was stable. Wireless, RADIUS, IoT, VoIP, VPN, SDN and QoS were integrated in stages. Each stage was tested before the next was added, and configuration changes were saved only after verification.

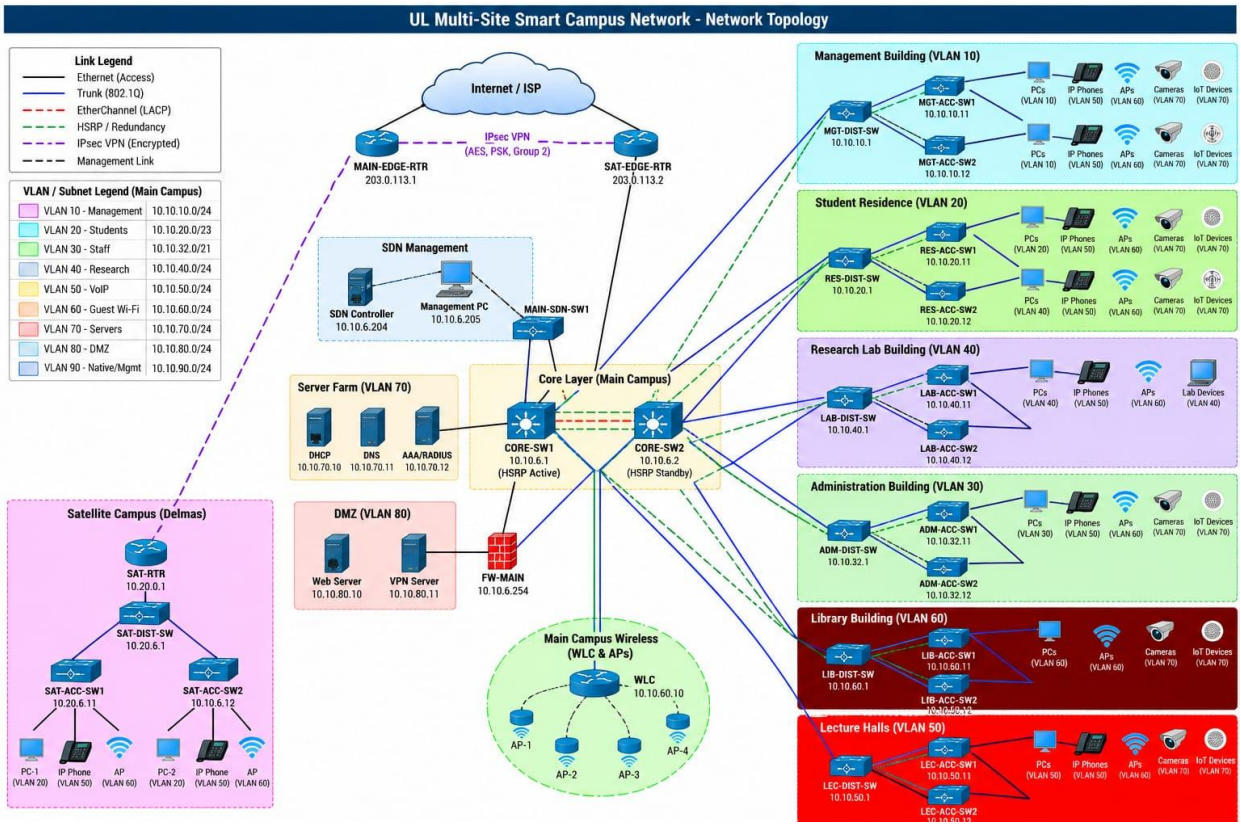


Figure 1 Complete multi-site smart campus topology

4 Network Architecture

4.1 Main Campus

The main campus follows a collapsed-core design. CORE-SW1 and CORE-SW2 perform inter-VLAN routing and HSRP. Each building switch has redundant trunks to the core, allowing spanning tree to select a forwarding path and retain an alternate path. MAIN-EDGE-RTR connects the campus to the simulated ISP and terminates the main side of the IPsec VPN. MAIN-SERVER-SW connects central services, the WLC, the ICT administration workstation and the ICT LAP.

4.2 Satellite Campus

The satellite campus uses SAT-CORE-SW for inter-VLAN routing, SAT-SW1 for user and wireless access, SAT-EDGE-RTR for WAN and VPN connectivity, and SAT-VOICE-RTR for local call control. A local WLC and LAP provide staff and student wireless services. The satellite retains local gateways and services while using the VPN to reach authorised central resources at the main campus.

4.3 Device Roles

Device or group	Primary role
CORE-SW1 / CORE-SW2	Main-campus Layer 3 switching, HSRP, ACLs and DHCP relay
MAIN-EDGE-RTR	WAN routing, zone firewall, IPsec VPN and QoS
SAT-CORE-SW	Satellite inter-VLAN routing and policy enforcement
SAT-EDGE-RTR	Satellite WAN routing, zone firewall, IPsec VPN and QoS
MAIN-VOICE-RTR / SAT-VOICE-RTR	Cisco CallManager Express and inter-campus dial peers
MAIN-WLC1 / SAT-WLC1	Central wireless LAN control and WLAN-to-VLAN mapping
Access switches	Endpoint VLAN assignment, trunking and port security
Central servers	DNS, DHCP, RADIUS, NTP/syslog, IoT, web and SDN services

5 IP Addressing and VLAN Design

5.1 Main Campus Addressing

VLAN	Name	Subnet	Virtual gateway	Purpose
10	ADMIN	10.10.7.0/25	10.10.7.1	Administrative users
20	STAFF	10.10.4.128/25	10.10.4.129	Academic and support staff

30	RESEARCH-LABS	10.10.7.128/25	10.10.7.129	Research laboratory computers
35	PRINTERS	10.10.6.0/26	10.10.6.1	Shared printers
40	STUDENTS	10.10.4.0/25	10.10.4.1	Wired student users
50	RESIDENCE	10.10.8.0/22	10.10.8.1	Residence students and residence WLAN
60	SERVERS	10.10.6.192/26	10.10.6.193	Central services and controllers
70	VOICE	10.10.6.64/26	10.10.6.65	IP phones and call control
80	IOT	10.10.5.128/25	10.10.5.129	Cameras and environmental sensors
90	STAFF-WIFI	10.10.5.0/25	10.10.5.1	Authenticated staff wireless
100	STUDENT-WIFI	10.10.0.0/[CONFIRM]	10.10.0.1	Authenticated student wireless
110	MANAGEMENT	10.10.6.128/26	10.10.6.129	Infrastructure management
120	ICT-ADMIN	10.10.12.0/27	10.10.12.1	Privileged ICT administration

5.2 Satellite Campus Addressing

VLAN	Name	Subnet	Gateway	Purpose
20	SAT-STAFF	10.20.20.0/24	10.20.20.1	Satellite staff
35	SAT-PRINTERS	10.20.35.0/24	10.20.35.1	Satellite printers
40	SAT-STUDENTS	10.20.40.0/24	10.20.40.1	Satellite students
70	SAT-VOICE	10.20.70.0/24	10.20.70.1	Satellite voice
80	SAT-IOT	10.20.80.0/24	10.20.80.1	Satellite IoT
90	SAT-STAFF-WIFI	10.20.90.0/24	10.20.90.1	Satellite staff wireless
100	SAT-STUDENT-WIFI	10.20.100.0/24	10.20.100.1	Satellite student wireless
110	SAT-MANAGEMENT	10.20.110.0/24	10.20.110.1	Satellite management

5.3 Addressing Rationale

VLSM was used at the main campus to match address capacity to expected demand. The residence and student wireless networks received larger allocations, while servers, printers, voice, management and ICT administration use smaller subnets. The satellite uses /24 networks to simplify local administration and allow predictable growth. Separate addressing also makes ACL rules, fault isolation and monitoring easier to understand.

6 Routing Redundancy and Layer 2 Resilience

6.1 Inter VLAN Routing

The main core switches use switched virtual interfaces and IP routing to route between VLANs. Access switches carry permitted VLANs across IEEE 802.1Q trunks. At the satellite campus, SAT-CORE-SW

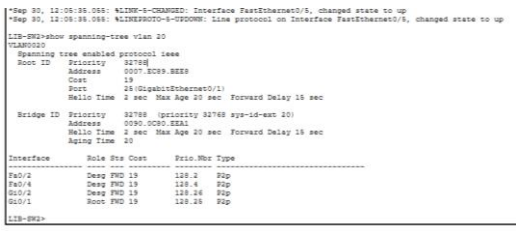
performs the same gateway function for the 10.20.0.0/16 site. DHCP relay addresses on SVIs forward client requests to the appropriate DHCP service.

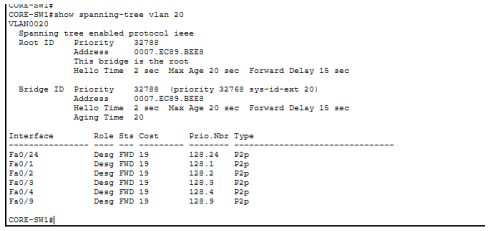
6.2 HSRP

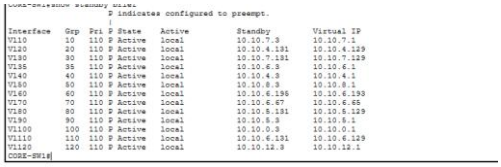
HSRP provides a stable virtual default gateway on the main campus. CORE-SW1 normally operates as Active with priority 110 and preemption enabled, while CORE-SW2 operates as Standby with priority 100. Failover testing showed CORE-SW2 becoming Active when communication with CORE-SW1 was interrupted. After restoration, the higher-priority core preempted and resumed the Active role. This was verified across all configured VLAN groups, including Admin, Staff, Research, Printers, Students, Residence, Servers, Voice, IoT, wireless, Management and ICT administration.

6.3 Spanning Tree and Redundant Uplinks

Per-VLAN spanning tree prevents loops created by dual access-switch uplinks. CORE-SW1 was assigned the lower bridge priority for key VLANs, with CORE-SW2 acting as the secondary path. During verification, one access uplink forwarded while the redundant link appeared as Alternate Blocking where necessary. Edge ports use PortFast, while BPDU Guard protects user-facing ports from accidental switch connections. LAP and WLC links use controlled trunk configurations and PortFast trunk where appropriate.







7 Security Architecture

7.1 Segmentation and Zero Trust Policy

The network applies least-privilege principles by placing different user and device categories in separate VLANs and applying extended ACLs at Layer 3 boundaries. Student and residence networks are prevented from initiating unrestricted access to internal campus networks. Research-laboratory users can reach authorised services but are isolated from unrelated user segments. IoT devices are restricted to their registration and management services. ICT administration receives broader access for authorised support tasks.

Control	Implementation and purpose
MAIN-STUDENT-ZERO-TRUST	Allows DHCP, DNS and approved services, then denies protected campus ranges before permitting external traffic

RESEARCH-POLICY	Allows central DNS and approved server access while denying lateral movement into user and management subnets
SERVER-ZERO-TRUST	Allows authorised management, RADIUS, IoT, NTP, syslog and service flows; denies other access to the server network
SAT-STUDENT-POLICY	Restricts satellite students while allowing required DNS and approved services
Zone-based firewall	Inspects authorised inside-to-outside sessions and drops unsolicited default traffic

7.2 Access Layer Hardening

- Unused switch ports were returned to default configuration, labelled UNUSED-SECURED-PORT and administratively shut down.
- User and IoT access ports use explicit access VLANs, PortFast and BPDU Guard where appropriate.
- Port security with restrict mode limits unauthorised MAC addresses on selected access ports.
- Infrastructure links use explicit trunk mode and restricted allowed VLAN lists.
- Device management uses local AAA and SSH from authorised management networks.

7.3 Central Authentication

The RADIUS server at 10.10.6.201 provides 802.1X authentication. Infrastructure devices use the configured shared secret and management source network. Wired research-laboratory ports were configured as authenticators with global 802.1X control. Wireless staff and student WLANs use enterprise authentication through the WLC. Testing showed that the Packet Tracer endpoint must have its 802.1X security option enabled before authentication succeeds.

7.4 Security Testing

Test	Expected result	Observed result	Status
Unauthorised student-to-protected network	Traffic denied by inbound ACL	Restricted traffic blocked; required DHCP/DNS retained	Pass
Unused switch port	No link or network access	Port remained administratively down	Pass
802.1X with incorrect credentials	Association or authorisation denied	Client did not obtain authorised access	Pass
802.1X with valid credentials	Authentication and correct VLAN service	Client connected and received DHCP service	Pass
VPN inspection	Active security associations and increasing counters	QM_IDLE and active IPsec SAs observed	Pass

8 Wireless and 802.1X Implementation

8.1 Wireless Architecture

Lightweight access points were deployed in the library, mathematics building, lecture halls, residences, ICT/server area and satellite campus. Each LAP joins its local controller through the native management VLAN. User WLAN traffic is mapped to the required staff, student or residence VLAN. Trunks between LAPs, WLCs and switches carry only the VLANs needed at each location.

Wireless service	SSID	VLAN	Authentication
Main staff	UL-STAFF-8021X	90	802.1X through RADIUS
Main students	UL-STUDENT-8021X	100	802.1X through RADIUS
Residence students	UL-RESIDENCE-8021X	50	802.1X through RADIUS
Satellite staff	UL-STAFF-8021X	90	802.1X through RADIUS
Satellite students	UL-STUDENT-8021X	100	802.1X through RADIUS

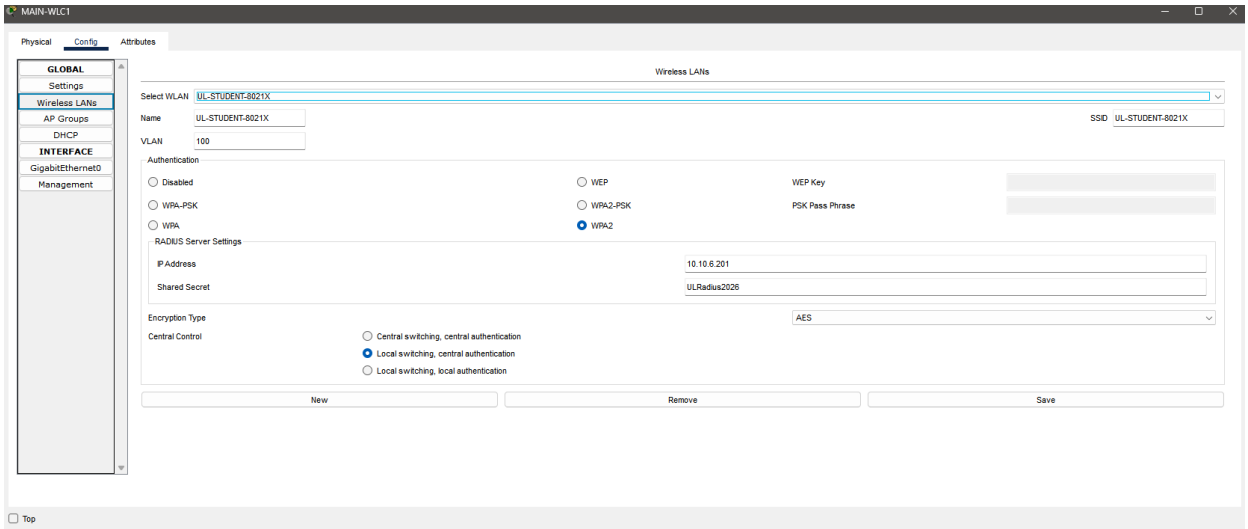
The main and satellite WLANs use the group name CAMPUS-USERS-8021X-GROUP to maintain consistent administration. Staff and student credentials are stored on the RADIUS service; passwords are not reproduced in this report.

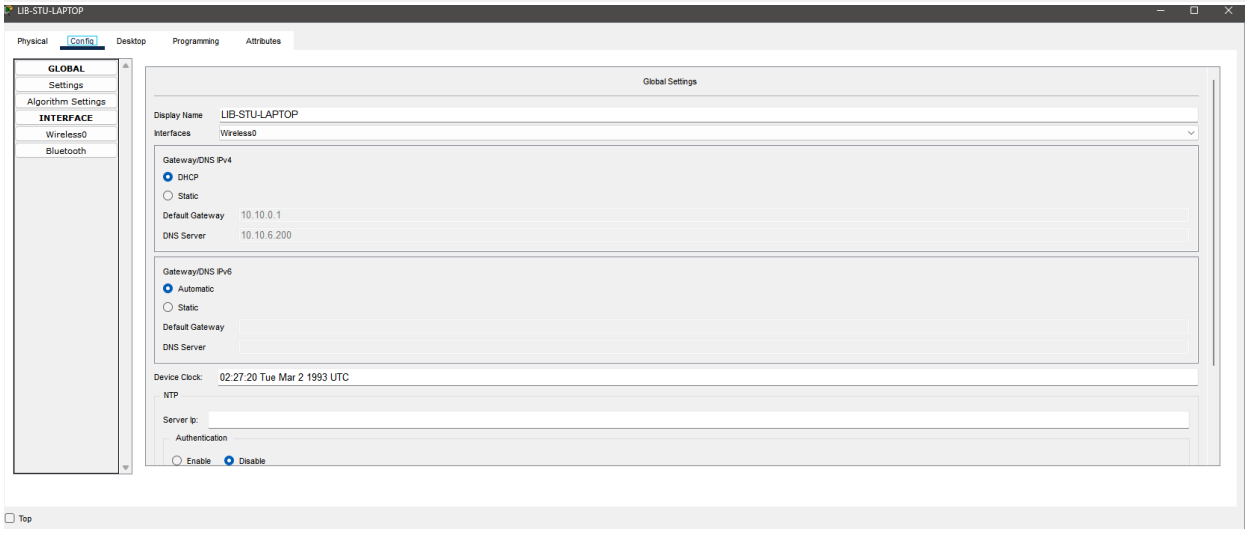
8.2 Residence Wireless Design

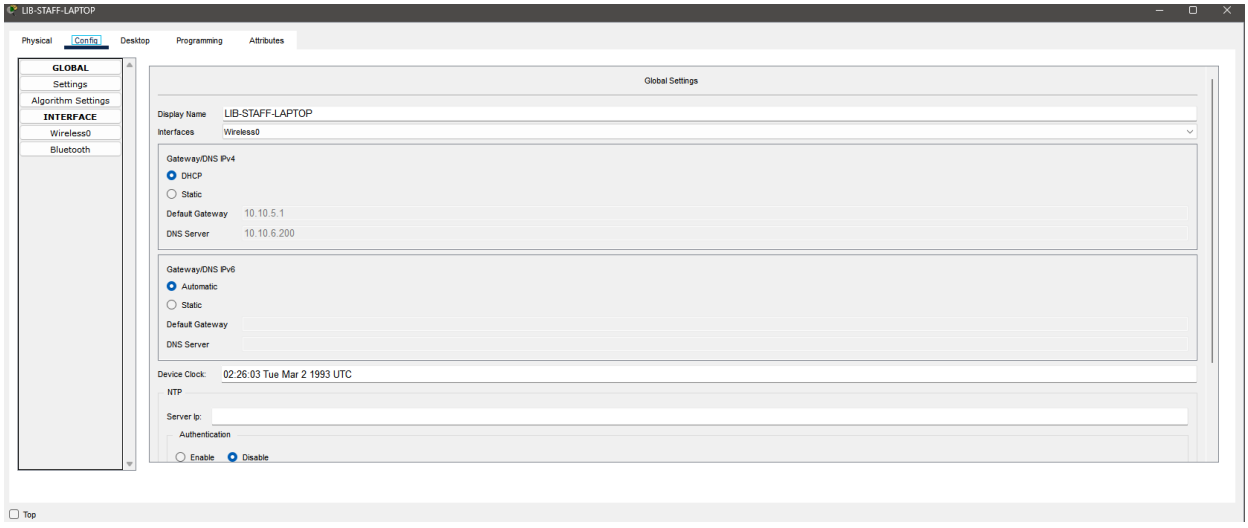
Residence users are separated from general student Wi-Fi because the residence network has different capacity and security requirements. Two LAPs were installed on each residence switch to improve coverage and support client density. Their access links carry VLAN 50 and VLAN 90 as required, with the native VLAN aligned to the LAP management design. Old autonomous student access points were removed and their switch ports were securely disabled.

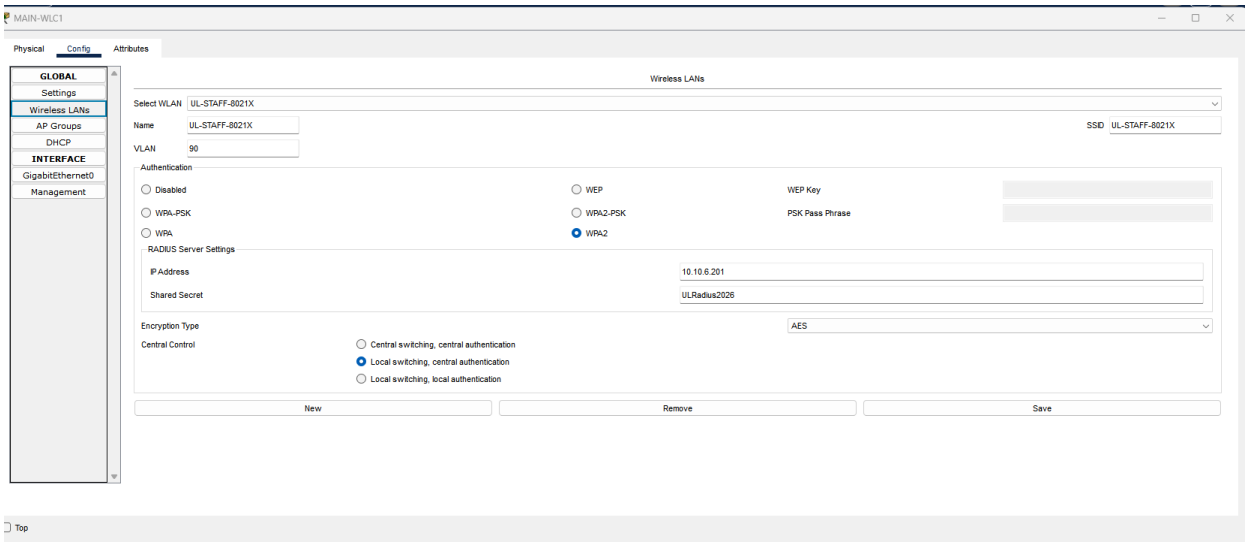
8.3 Wireless Troubleshooting

Several LAPs initially remained offline or took an extended period to join. Troubleshooting checked switch-port status, MAC learning, native VLAN consistency, allowed VLANs, spanning-tree forwarding and DHCP availability. Restarting Packet Tracer was occasionally required because the simulator retained stale wireless state. New clients also required SSID reselection and DHCP renewal after authentication. The final test confirmed successful LAP registration and address allocation for staff, students and residence users.


MAIN-WLC1
Physical
Config
Attributes
GLOBAL
Settings
Wireless LANs
AP Groups
DHCP
INTERFACE
GigabitEthernet0
Management
Wireless LANs
Select WLAN
UL-STUDENT-8021X
Name
UL-STUDENT-8021X
SSID
UL-STUDENT-8021X
VLAN
100
Authentication
Disabled
WEP
WEP Key
WPA-PSK
WPA2-PSK
PSK Pass Phrase
WPA
WPA2
RADIUS Server Settings
IP Address
10.10.6.201
Shared Secret
ULRadius2026
Encryption Type
AES
Central Control
Central switching, central authentication
Local switching, central authentication
Local switching, local authentication
New
Remove
Save
Top


LIB-STU-LAPTOP
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Wireless0
Bluetooth
Global Settings
Display Name
LIB-STU-LAPTOP
Interfaces
Wireless0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
10.10.0.1
DNS Server
10.10.6.200
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock:
02:27:20 Tue Mar 2 1993 UTC
NTP
Server Ip:
Authentication
Enable
Disable
Top


LIB-STAFF-LAPTOP
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
Wireless0
Bluetooth
Global Settings
Display Name
LIB-STAFF-LAPTOP
Interfaces
Wireless0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
10.10.5.1
DNS Server
10.10.6.200
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Device Clock:
02:26:03 Tue Mar 2 1993 UTC
NTP
Server Ip:
Authentication
Enable
Disable
Top



9 VPN and Multi Site Connectivity

MAIN-EDGE-RTR and SAT-EDGE-RTR establish a site-to-site IPsec VPN across the simulated ISP. The protected networks are 10.10.0.0/16 and 10.20.0.0/16. IKE negotiation reached QM_IDLE and the IPsec security associations showed active inbound and outbound ESP SAs. Encryption and decryption counters increased on both peers, confirming two-way protected traffic rather than a configuration-only tunnel.

VPN component	Main campus	Satellite campus
Public endpoint	203.0.113.1	198.51.100.2
Protected network	10.10.0.0/16	10.20.0.0/16
Crypto map	UL-VPN	UL-VPN
Observed state	QM_IDLE / ACTIVE	QM_IDLE / ACTIVE
Traffic evidence	Encapsulation and decapsulation counters increased	Encapsulation and decapsulation counters increased

```
Password:
SAT-EDGE-RTR#enable
SAT-EDGE-RTR#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
203.0.113.1     198.51.100.2    QM_IDLE           1097    0 ACTIVE


IPv6 Crypto ISAKMP SA


SAT-EDGE-RTR#show crypto ipsec sa

interface: FastEthernet0/1
    Crypto map tag: UL-VPN, local addr 198.51.100.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.20.0.0/255.255.0.0/0/0)
   remote  ident (addr/mask/prot/port): (10.10.0.0/255.255.0.0/0/0)
   current_peer 203.0.113.1 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 3906, #pkts encrypt: 3906, #pkts digest: 0
   #pkts decaps: 4055, #pkts decrypt: 4055, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 1, #recv errors 0

     local crypto endpt.: 198.51.100.2, remote crypto endpt.:203.0.113.1
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/1
     current outbound spi: 0x13081CC7(319298759)

     inbound esp sas:
      spi: 0x176D58B5(393042101)

SAT-EDGE-RTR#ping 10.10.x.x
Translating "10.10.x.x"
% Unrecognized host or address or protocol not running.
```

```
Password:
MAIN-EDGE-RTR#enable
MAIN-EDGE-RTR#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
198.51.100.2    203.0.113.1     QM_IDLE           1003    0 ACTIVE


IPv6 Crypto ISAKMP SA


MAIN-EDGE-RTR#show crypto ipsec sa

interface: FastEthernet1/0
    Crypto map tag: UL-VPN, local addr 203.0.113.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.10.0.0/255.255.0.0/0/0)
   remote  ident (addr/mask/prot/port): (10.20.0.0/255.255.0.0/0/0)
   current_peer 198.51.100.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 4008, #pkts encrypt: 4008, #pkts digest: 0
   #pkts decaps: 3846, #pkts decrypt: 3846, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 203.0.113.1, remote crypto endpt.:198.51.100.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet1/0
     current outbound spi: 0x176D58B5(393042101)

     inbound esp sas:
```

10 VoIP and Quality of Service

10.1 Voice Services

Cisco CallManager Express was configured on MAIN-VOICE-RTR and SAT-VOICE-RTR. Main-campus extensions use the 2000 range and satellite extensions use the 3000 range. DHCP option 150 directs phones to the correct local call-control address. Registered ephones displayed their extensions and remained reachable through their local voice VLANs. Dial peers route 3xxx calls from the main campus to the satellite voice gateway and 2xxx calls from the satellite to the main voice gateway.

Site	Call-control address	Extension range	Inter-site dial peer
Main campus	10.10.6.68:2000	2001-2009	3... to 10.20.70.2
Satellite campus	10.20.70.2	3001 onward	2... to 10.10.6.68

A live inter-campus call between extensions 3001 and 2004 displayed Connected, proving dial-plan and VPN reachability.

10.2 QoS Design

WAN QoS is applied outbound on MAIN-EDGE-RTR FastEthernet1/0 and SAT-EDGE-RTR FastEthernet0/1. Voice receives a strict-priority queue equal to 20 percent of interface bandwidth. The main campus also reserves 30 percent for video. Voice traffic is identified on the inside interfaces and marked DSCP EF before encryption. The WAN voice classes match both the voice ACL and DSCP EF.

Router	Policy	Voice treatment	Additional treatment
MAIN-EDGE-RTR	MAIN-WAN-QOS	Strict priority 20%	Video bandwidth 30%
SAT-EDGE-RTR	SAT-WAN-QOS	Strict priority 20%	Fair queue for default traffic

Packet Tracer does not support the qos pre-classify command under the crypto map and does not fully reproduce DSCP propagation through its simulated IPsec process. As a result, the inside marking policies recorded voice matches while WAN priority counters remained very low. The policy attachment, traffic classification, DSCP marking and successful call were retained as the technically correct design; prioritising all ESP traffic was rejected because it would incorrectly treat every VPN application as voice.

```
MAIN-VOICE-RTR#show ephone

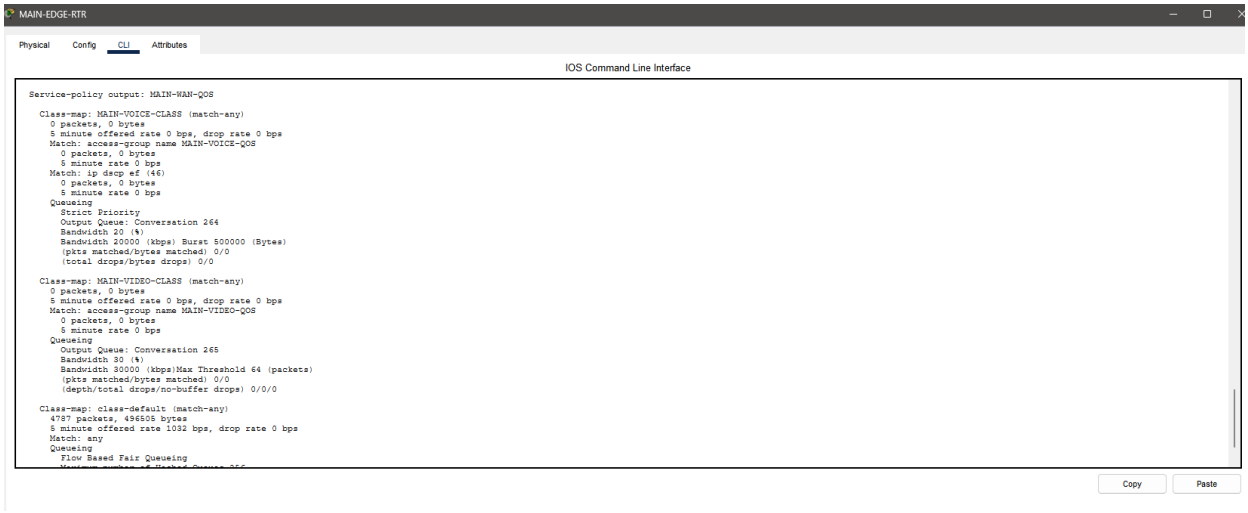
ephone-1 Mac:00E0.F740.D2D2 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.91 1026 7960   keepalive 43 max_line 2
 button 1: dn 1  number 2001 CH1   IDLE

ephone-2 Mac:0060.2F20.7935 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.83 1026 7960   keepalive 43 max_line 2
 button 1: dn 5  number 2005 CH1   IDLE

ephone-3 Mac:0001.96BA.D57C TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.84 1026 7960   keepalive 43 max_line 2
 button 1: dn 4  number 2004 CH1   IDLE

ephone-4 Mac:00D0.D3AC.6BD8 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.85 1026 7960   keepalive 43 max_line 2
 button 1: dn 2  number 2002 CH1   IDLE

ephone-5 Mac:000A.F399.1A92 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
 --More--
```



11 IoT and Central Services

11.1 Central Services

Service	Address	Function
DNS and DHCP	10.10.6.200	Name resolution and central address allocation
RADIUS	10.10.6.201	802.1X authentication
IoT registration	10.10.6.202	Registers and manages smart devices
NTP and Syslog	10.10.6.203	Time synchronisation and central logging
SDN controller	10.10.6.204	Network discovery and programmable monitoring
Main WLC	10.10.5.10	Main-campus wireless control

11.2 IoT Segmentation

Cameras, smoke detectors, temperature sensors and related devices use dedicated IoT VLANs. Main-campus IoT devices use VLAN 80 and the 10.10.5.128/25 network, while satellite devices use 10.20.80.0/24. ACLs permit registration with the IoT server while reducing access to unrelated campus

systems. Registration was verified from the ICT administration workstation through the IoT server browser interface.

11.3 Monitoring

Network devices synchronise time with the NTP service and forward supported log events to the central syslog server. Consistent timestamps improve incident investigation and configuration auditing. The SDN monitoring script authenticates to the controller API, requests the managed-device inventory and reports hostname, management address, device type, reachability and collection status.

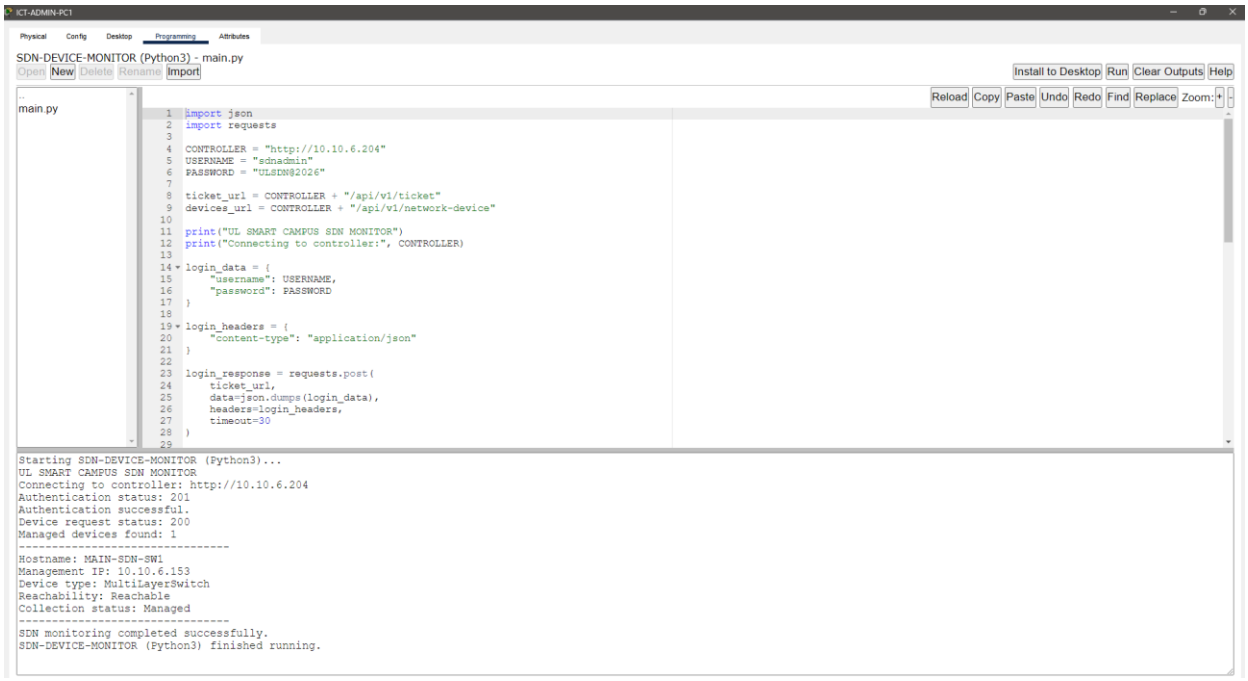
12 SDN Integration and Automation

12.1 SDN Controller

The controller at 10.10.6.204 is located in the protected server network. The ICT administration workstation at 10.10.12.15 can reach the controller through the permitted management path. The controller provides central visibility of managed infrastructure and exposes REST endpoints for basic programmability.

12.2 Monitoring Script

A Python script running from the designated programming workstation authenticates to the controller ticket endpoint and retrieves the network-device inventory. This demonstrates separation between the programming workstation and the ICT administration workstation used for browser-based controller management. Authentication secrets are intentionally omitted from the report and should be stored outside source code in a production implementation.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfigping 10.10.6.204
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::260:70FF:FE7B:3A23
   IPv6 Address.....: ::
   IPv4 Address.....: 10.10.12.11
   Subnet Mask.....: 255.255.255.224
   Default Gateway.....: ::
                                 10.10.12.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                                 0.0.0.0

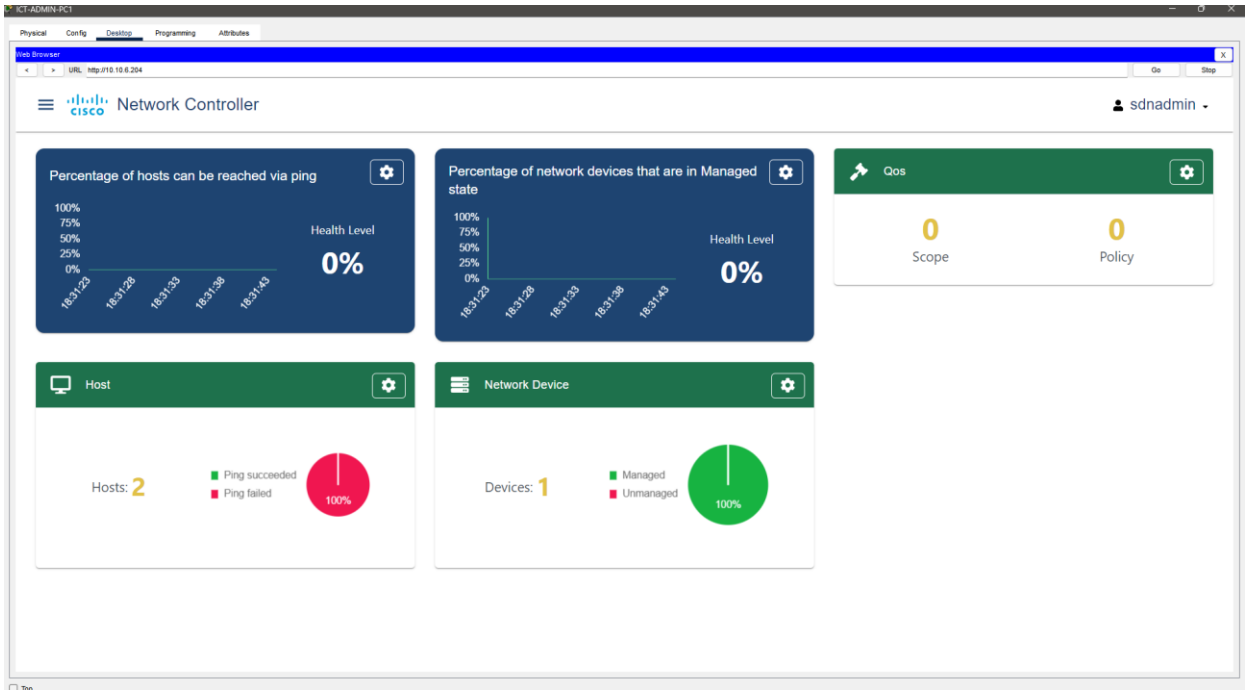
C:\>ping 10.10.6.204

Pinging 10.10.6.204 with 32 bytes of data:

Reply from 10.10.6.204: bytes=32 time<1ms TTL=127
Reply from 10.10.6.204: bytes=32 time=11ms TTL=127
Reply from 10.10.6.204: bytes=32 time<1ms TTL=127
Reply from 10.10.6.204: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.6.204:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```



12.3 Recommended Automation Extension

A future phase should export device configurations on a schedule and store them in a controlled Git repository. Ansible or Python/Netmiko could validate VLAN, trunk, NTP, syslog and SSH settings against a standard baseline. Credentials should be provided through environment variables or a secrets vault rather than committed to the repository.

13 Testing Troubleshooting and Performance Analysis

13.1 Final Test Plan and Results

ID	Test	Method	Observed result	Status
T01	DHCP allocation	Renew clients in representative VLANs	Correct subnet, gateway and DNS assigned after simulator convergence	Pass
T02	Inter-VLAN routing	Ping permitted gateways and services	Permitted destinations reachable	Pass
T03	Student isolation	Attempt access to protected internal ranges	ACL denied prohibited traffic	Pass
T04	HSRP failover	Interrupt active-core path and observe states	Standby became Active; preferred core recovered with preemption	Pass
T05	STP redundancy	Inspect forwarding and alternate uplinks	Loop-free topology with alternate path retained	Pass
T06	VPN establishment	Inspect IKE and IPsec SAs	QM_IDLE and active bidirectional SAs	Pass

T07	Wireless 802.1X	Connect with valid and invalid accounts	Valid account connected; invalid account denied	Pass
T08	Residence Wi-Fi	Join residence SSID and request DHCP	Client received a 10.10.8.0/22 address	Pass
T09	VoIP registration	Inspect ephones at both sites	Phones registered with correct extensions	Pass
T10	Inter-campus call	Call 3001 to 2004	Call displayed Connected	Pass
T11	QoS policy	Inspect marking and WAN policies	Voice identified/marked; strict priority policies attached	Pass with simulator limitation
T12	IoT registration	Use ICT browser to inspect registration service	Authorised IoT devices visible	Pass
T13	SSH management	Ping and SSH to managed device	Reachability confirmed; SSH enabled where configured	Pass
T14	NTP and logging	Inspect clock and server configuration	Devices reference 10.10.6.203	Pass

13.2 Troubleshooting Record

Problem	Root cause	Corrective action	Outcome
VLAN absent from forwarding list	VLAN not created or STP still converging	Create VLAN, permit it end-to-end and wait for forwarding	Traffic restored
Residence LAP offline	Native VLAN inconsistency or stale simulator state	Verify trunk/native VLAN, MAC learning, DHCP and restart Packet Tracer	LAP joined controller
Wireless client had no address	SSID/VLAN mapping or delayed DHCP state	Correct WLAN mapping, reselect SSID and renew DHCP	Valid address obtained
Research 802.1X failed	Endpoint 802.1X option disabled	Enable 802.1X security and retry credentials	Authentication succeeded
RADIUS ping returned unreachable	Access policy or source-path behaviour	Verify management SVI, ACL permission and RADIUS settings	Authentication traffic permitted
HSRP dual-active event	Temporary loss of hello communication during topology changes	Restore VLAN path and verify STP/trunks	Normal Active/Standby state restored
WAN voice counters remained low	Packet Tracer IPsec and RTP simulation limitation	Mark DSCP EF before encryption and document limitation	Correct policy retained

13.3 Performance Analysis

Campus pings generally completed with low simulated delay. A representative test to the main WLC returned four of five replies after initial ARP resolution, with subsequent communication stabilising. VPN packet counters increased in both directions without receive errors. HSRP restored gateway availability after convergence, while spanning tree maintained loop-free redundant paths. No QoS drops were observed because the simulated load was substantially below interface capacity.

The results indicate that segmentation and redundancy goals were achieved. The design is capable of supporting the simulated population, but a production deployment would require wireless site surveys, measured WAN bandwidth, realistic concurrent traffic generation, vulnerability scanning and hardware performance baselines.

WAN QoS policy and traffic counters

```
MAIN-EDGE-RTR#show policy-map interface fastEthernet0/1
 FastEthernet0/1

  Service-policy input: MAIN-VOICE-MARK

    Class-map: MAIN-VOICE-CLASS (match-any)
      0 packets, 0 bytes
      5 minute offered rate 0 bps, drop rate 0 bps
      Match: access-group name MAIN-VOICE-QOS
        0 packets, 0 bytes
        5 minute rate 0 bps
      Match: ip dscp ef (46)
        0 packets, 0 bytes
        5 minute rate 0 bps

    Class-map: class-default (match-any)
      58 packets, 14335 bytes
      5 minute offered rate 55 bps, drop rate 0 bps
      Match: any

MAIN-EDGE-RTR#
```

Copy Paste

Active site-to-site IPsec VPN tunnel

```
CORE-SW1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

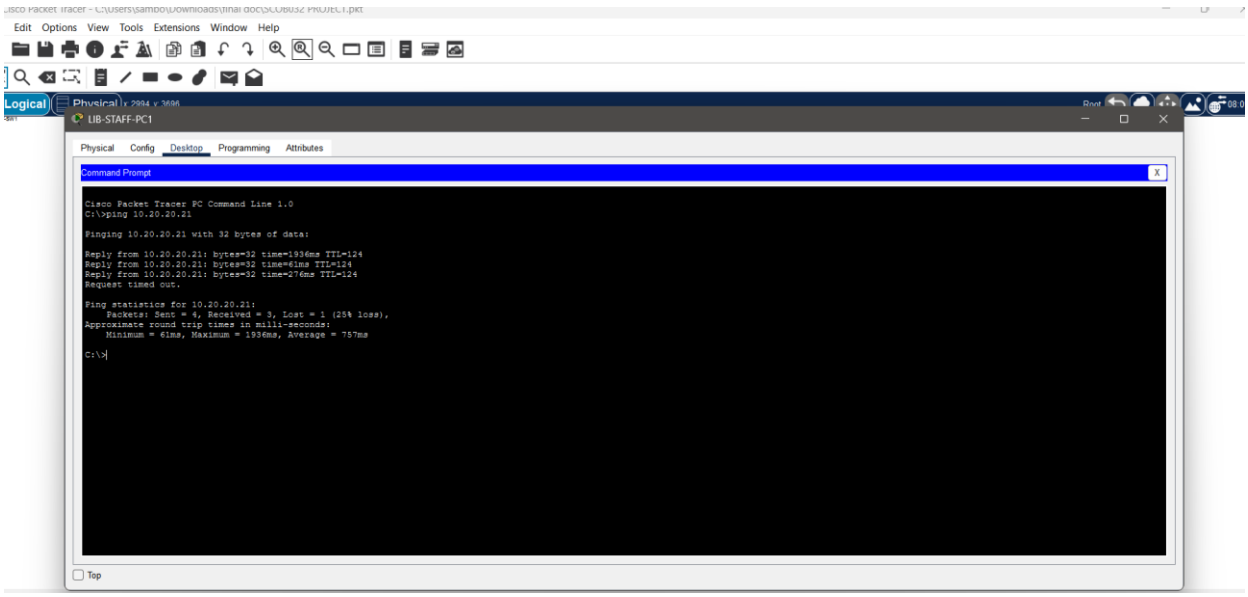
IPv6 Crypto ISAKMP SA
```

HSRP gateway redundancy and failover

```
CORE-SW1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           10.10.7.3       10.10.7.1
Vl120       20   110 P Active   local           10.10.4.131     10.10.4.129
Vl130       30   110 P Active   local           10.10.7.131     10.10.7.129
Vl135       35   110 P Active   local           10.10.6.3       10.10.6.1
Vl140       40   110 P Active   local           10.10.4.3       10.10.4.1
Vl150       50   110 P Active   local           10.10.8.3       10.10.8.1
Vl160       60   110 P Active   local           10.10.6.195     10.10.6.193
Vl170       70   110 P Active   local           10.10.6.67      10.10.6.65
Vl180       80   110 P Active   local           10.10.5.131     10.10.5.129
Vl190       90   110 P Active   local           10.10.5.3       10.10.5.1
Vl1100      100  110 P Active   local           10.10.0.3       10.10.0.1
Vl1110      110  110 P Active   local           10.10.6.131     10.10.6.129
Vl1120      120  110 P Active   local           10.10.12.3      10.10.12.1
CORE-SW1#
```

```
CORE-SW2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100 P Speak    10.10.7.3       unknown         10.10.7.1
Vl120       20   100 P Standby  10.10.4.130     local           10.10.4.129
Vl130       30   100 P Standby  10.10.7.130     local           10.10.7.129
Vl135       35   100 P Standby  10.10.6.2       local           10.10.6.1
Vl140       40   100 P Standby  10.10.4.2       local           10.10.4.1
Vl150       50   100 P Standby  10.10.8.2       local           10.10.8.1
Vl160       60   100 P Standby  10.10.6.194     local           10.10.6.193
Vl170       70   100 P Standby  10.10.6.66      local           10.10.6.65
Vl180       80   100 P Standby  10.10.5.130     local           10.10.5.129
Vl190       90   100 P Standby  10.10.5.2       local           10.10.5.1
Vl1100      100  100 P Standby  10.10.0.2       local           10.10.0.1
Vl1110      110  100 P Speak    10.10.6.131     unknown         10.10.6.129
Vl1120      120  100 P Standby  10.10.12.2      local           10.10.12.1
CORE-SW2#
```

Successful inter-campus connectivity test



14 Limitations and Recommendations

14.1 Simulation Limitations

- Packet Tracer does not implement every Cisco IOS command or protocol feature.
- The simulator does not reproduce real wireless propagation, interference or roaming behaviour.
- IPsec QoS pre-classification and continuous RTP generation are limited.
- Performance measurements do not represent physical link utilisation, jitter or hardware forwarding capacity.
- The controller and IoT implementations expose a simplified subset of production features.

14.2 Recommendations

- Validate the design in GNS3, EVE-NG or a physical Cisco lab for advanced firewall and QoS behaviour.
- Use dynamic routing with authentication where the production topology grows beyond the current static design.
- Deploy redundant wireless controllers and complete formal RF site surveys.
- Use certificate-based EAP-TLS instead of shared-password authentication for production wireless access.
- Add central SIEM integration, vulnerability scanning, configuration compliance checks and alerting.
- Store RADIUS, controller and device credentials in a protected secrets-management platform.
- Measure throughput, latency, jitter and packet loss under realistic concurrent voice, video and data loads.
- Maintain tested configuration backups and a controlled change-management process.

15 Conclusion

The project produced a functional and secure multi-site smart campus prototype that meets the main design objectives. The main and satellite campuses are connected through an active IPsec VPN, users and services are segmented through VLANs and ACLs, and redundant Layer 2 and Layer 3 mechanisms improve availability. Central DHCP, DNS, RADIUS, monitoring, IoT and SDN services support consistent operation across the environment.

The implementation also demonstrates enterprise wireless authentication, registered lightweight access points, inter-campus VoIP and QoS policy design. Troubleshooting revealed practical dependencies between VLAN creation, trunk permissions, spanning tree, DHCP, authentication and application services. These findings show that successful campus networking depends on testing the complete traffic path rather than treating each device configuration in isolation.

Within Packet Tracer's limitations, the final prototype provides a defensible foundation for a scalable University of Limpopo smart campus network. The next stage should reproduce the design in a more feature-complete emulator or physical lab and add production-grade identity, monitoring and performance testing.

16 Individual Contribution Statement

Complete this table accurately before submission. Contributions should describe work actually performed, such as addressing, switching, routing, security, wireless, servers, testing, documentation or presentation preparation.

Member name	Student number	Responsibilities and deliverables
READANI NF	2023057803	DHCP SETUP
RAKGETSE PI	202204262	VPN SETUP
RAMAHLO RS	202391068	QoS RESEARCHER
RAMOLOTO CB	240000760	FIREWALL CONFIGURATION
RAMPHANDI KM	240036726	TOPOLOGY DESIGN
RANHLAKGWE MM	202379376	VLAN CONFIGURATION
RAPOTU K	202173081	DNS CONFIGURATION
RASIVHETSHELE M	240040718	IOT CONFIGURATION
RAVHURA TC	202322062	WIRELESS 802.1 SETUP
SAMBO NL	202409799	SUBNET PLANNER
SEKGOBELA TM	240015914	VLAN CONFIGURATION
SEKOTO B	202103276	VLM PLANNER
SEKUBA MM	240036268	IOT CONFIGURATION
SEMENYA M	202374221	DOCUMENTATION

17 References

1. Cisco Systems. Cisco IOS Security Configuration Guide: Secure Connectivity and VPN.
2. Cisco Systems. Catalyst Switch Configuration Guides: VLANs, STP, HSRP, Port Security and 802.1X.
3. Cisco Systems. Enterprise QoS Solution Reference Network Design Guide.
4. IEEE. IEEE Std 802.1X: Port-Based Network Access Control.
5. National Institute of Standards and Technology. SP 800-207: Zero Trust Architecture.
6. University of Limpopo Department of Computer Science. SCOB032 Computer Networks Semester Project Brief, 2026.

Appendix A Key Verification Commands

- `show interfaces status`
- `show interfaces trunk`
- `show vlan brief`
- `show spanning-tree vlan <vlan-id>`
- `show standby brief`
- `show ip route`
- `show access-lists`
- `show crypto isakmp sa`
- `show crypto ipsec sa`
- `show policy-map interface <interface>`
- `show ephone`
- `show ip interface brief`
- `show mac address-table interface <interface>`
- `show running-config | section <section>`

Appendix B Evidence Register

Evidence ID	Screenshot or output	Report section
E01	Complete physical topology	3 and 4
E02	VLAN and trunk verification	5 and 6
E03	HSRP Active/Standby states	6
E04	STP root and blocked alternate path	6
E05	ACL counters and isolation test	7
E06	WLC LAP and WLAN status	8
E07	Wireless client DHCP and authentication	8
E08	IKE and IPsec security associations	9
E09	Registered ephones and connected call	10
E10	QoS marking and WAN policies	10
E11	IoT registration page	11
E12	SDN controller inventory and script output	12

E13	DHCP, ping and traceroute results	13